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EDUCATION & SKILLS

Rice University, Houston, TX May 2024
B.S. Mechanical Engineering - Specialization in Thermal Fluids GPA: 3.7/4.0
Certificate in Engineering Leadership (RCEL)

Relevant Coursework: Numerical Methods for PDEs, FEA, CFD, Control System Analysis, LHPT

Skills: Languages: Python, C++, C | **Controls:** PLC, SCADA, MES, Arduino | **Tools:** SolidWorks (CSWP-S), LabVIEW, Git | **PM:** WBS, OKR Facilitation| **Networking:** OPCUA, Cisco, TCP/IP

EXPERIENCE & PROJECTS

Dynamic Motion Control Inc. Systems Engineer, September 2024 – Present

- Delivered 800+ hours of client-facing integration engineering across 30+ clients spanning Rare Minerals, Metallurgy, and Defense; earned 100% CSAT on all projects with PM responsibilities.
- Led Factory Acceptance Testing for a Rocket Engine Static Test Fire Stand and Explosives Mixing Facility at Navy Air Weapons Test Facilities; programmed in new tech stacks under aggressive timelines including: NI Hardware, Cisco, Rockwell, OEM hardware, LabVIEW, and Python.
- Produced IO documentation for 1,500+ control system points aboard the MV Kennicott, serving as the functional basis for the vessel's control system FDS where no prior documentation existed.
- Authored company's first OKRs tailored to meet best practice and objectives; Led multiple office presentations on OKR use and implementation; Coached VPs; Drove 100% review adoption.
- Automated renaming of 19,000 unnamed variables across a 6-PLC system by building a bidirectional dictionary function which parsed XML exports eliminating thousands of hours of manual work.

Senior Capstone: Sea++, Autonomous Catamaran Team Lead, August 2023 – May 2024

- Elected by 7 peers to lead the design and construction of a 9-foot autonomous catamaran.
- Project commissioned to decentralize transportation of goods and failure points for Navy with the long-term goal of mass manufacturing at low cost (less than $7500 per boat).
- Achieved multipoint autonomous navigation in 5-ft seas and 25-mph winds with a 300-lb load, demonstrating integration of local and global control using Modified A*, Adaptive LOS, and PID.
- Designed the wing and hull with CFD analysis. Built with laser cut ribs, XPS foam, and fiberglass.

Rice Robotics: President & Manufacturing Lead, April 2023 – May 2024

- Grew club 18x from 5 to 90+ active members through: organizational restructuring, project leadership, and strategic advertising. Technical lead for Robotic Cameraman project (RoboCam).

Fernware: Automatic Home Plant Watering System, August 2025 – Present

- Friends killed my plants twice. Built a modular automatic watering system with moisture sensors.

Bay Head Yacht Club: C420 Advanced Sailing Instructor, May 2018 – August 2022

- Coached for 5 years. Top pupils placed 3rd and 7th in C420 New Jersey State Championship 2022

HOBBIES & INTERESTS

Ironman Triathlons; Bread baking; Cheese making; Cook; Surfer and shaper; Classical Pianist 10+ yrs; Electric guitar; Self-taught kiteboarder; Beach Volleyball; Chess (1600 Rapid); Strategy board games; Voracious Reader; Intermediate Spanish; Intermediate Mandarin; World Traveling.